3D Printing PDB Structures

Summary:

For generating a 3D printable model of your structure, you will need a PDB file. If this is structure is an assembly (e.g., icosahedral virus) then apply the necessary symmetry operation to generate the entire assembly. For instructions on how to do this please go to the Hormel Institute Wiki. Also, you will need to use UCSF ChimeraX. To download, please go to the ChimeraX Download Page.

Procedure:

1. Setting up ChimeraX

a. First download ChimeraX and install it on your computer.
b. Then go to the 'Tools' menu and select 'More Tools'.
c. In the small pop-up window search for the 'NIHPresets' app and install it.

2. Preparing your PDB file

a. First open your PDB file in ChimeraX.
b. If your PDB file is an icosahedral virus, first generate the entire assembly:

```
sym [model #] I,222
```

c. Now go to the 'presets' menu and select 'NIH3D'.
d. Select the best preset for your model. Make sure that the preset says '(Printable)' at the end.
e. Now issue the command:

```
surface #1 resolution 5 gridSpacing 2
```

f. You can verify the number of triangles with the 'graphics triangles'' command.

3. Saving your print file

a. You'll want to save your print file in one of two formats:
   i. OBJ (Wavefront Object) files store geometry data and can also store texture and material information. They are widely readable by many slicing programs, though they are often larger files.
   ii. GLB (GL Transmission Format Binary) files are basically OBJ files, but in a compressed binary format. They are typically smaller than OBJ files, however, GLB files are not as commonly supported by 3D printing software.